

Chapter 1: Introduction

1.1 Problem Statement

Marine radar systems struggle to detect very small, low radar-cross-section (RCS) targets (i.e. partially submerged swimmers, floating buoys, or debris) against the dynamic ocean background.[1,3,4] These objects often produce weak radar echoes that are easily masked by sea clutter (the radar returns from waves).[2,3,18] For example, maritime surveillance radars have difficulty reliably picking up small craft or persons in the water because the sea clutter is large and the targets' RCS is small, reducing the probability of detection[19,1]. Even specialized periscope-detection radars on patrol aircraft encounter significant challenges in distinguishing such small targets from wave reflections, leading to high false-alarm rates in rough sea conditions[20,7]. In short, conventional marine radars are limited in detecting swimmers, small buoys, and similar low-RCS objects, especially amid heavy sea clutter and wave interference.

At the same time, developing robust vision-based detectors for maritime environments is hampered by a shortage of annotated training data[8,9]. Curating labeled image datasets of rare small objects at sea is expensive and time-consuming[10,11], and the few public maritime image datasets that do exist are limited in scale and diversity[12,13]. The result is a data bottleneck; deep learning models demand large numbers of labeled examples to achieve high accuracy[14], yet acquiring and labeling such examples on the open ocean is impractical[15]. This labeling bottleneck is especially acute for maritime imagery, where expert annotators must sift through vast scenes of mostly water to mark tiny, distant objects[16]; a process far more costly and difficult than labeling common objects in terrestrial scenes[17].

However, radar's all-weather sensing capabilities present an opportunity to mitigate this problem.[21,22] Radar can detect objects at long range and under poor visibility conditions (night, fog, rain) where optical cameras often fail[6,23,24]. By using radar detections to cue or confirm the presence of objects in camera images, we can effectively "seed" the labeling process. For instance, a radar return indicating a target can flag a region of interest in a simultaneously captured camera frame, prompting a human annotator (or an assisting algorithm) to verify and label the object in the optical image. Researchers have begun exploring such sensor cueing approaches; for example, a Radar Region Proposal Network (RRPN) was proposed to project radar detections into the camera view as candidate object regions[6,25,26]. This concept suggests that radar could serve as an automatic region proposal generator for images, focusing human attention on likely object locations; especially valuable when the objects are subtle or the visibility is poor.

The motivation for this thesis is therefore to bridge the radar and optical modalities to overcome their individual limitations. A fusion of radar and camera information promises more robust detection of small maritime objects and a means to automate data labeling. Radar can compensate for the camera's shortcomings in adverse conditions, while the camera provides the high-resolution classification that radar lacks. By jointly using radar and optical sensors, we aim to improve detection performance for small objects in the maritime domain and to develop an automated labeling aid, thereby addressing both the detection gap of radar and the data scarcity problem of deep learning models.

1.2 Research Objectives

Based on the above problem, the objectives of this research are as follows:

- (1) Model the clutter and background; develop a modeling framework for sea clutter and sensor noise to better distinguish small targets from background returns. This will involve characterizing the statistical properties of sea clutter (which can produce spurious spikes) in various conditions so as to minimize false alarms and missed detections.
- (2) Optimize radar detectability; optimize the radar's signal processing settings (such as gain, pulse parameters, and detection thresholds) to maximize the detectability of low-RCS objects. In particular, adapt the radar's automatic gain control or CFAR thresholding across each sweep to maintain a constant false-alarm rate while boosting sensitivity to small targets.
- (3) Co-register radar and camera data; establish an accurate calibration and co-registration between the radar and camera sensors. This objective entails determining the precise spatial and temporal alignment so that radar detections can be correctly mapped onto the camera image frame.
- (4) Automate label generation with radar cues; implement a radar-assisted labeling pipeline in which radar detections provide initial regions of interest (ROIs) for objects in the camera images. The goal is to use radar returns as prompts for where a potential object might be, thereby automating part of the image annotation process and reducing the manual effort.
- (5) Evaluate fused detection performance; rigorously evaluate the performance of the proposed radar-camera fusion approach for small-object detection and the effectiveness of the automated labeling. This involves testing under representative maritime scenarios (different sea states, lighting, and weather conditions) and comparing against baseline single-sensor detectors. Key metrics will include detection probability, false-alarm rate, localization accuracy, and the efficiency and quality of the generated labels.

1.3 Project Context

1.3.1 Marine Radar for Small-Object Detection

1.3.1.1 Radar operating modes and bands

Marine radars typically operate in either pulsed or frequency-modulated continuous wave (FMCW) modes[27,28,29,30]. Traditional pulse radars emit high-power RF pulses and listen for echoes during the intervals between pulses. One drawback of pulse radar is a *minimum range blind spot*; effectively, there is a short distance (on the order of tens of meters) in front of the radar within which it cannot receive reflections immediately after transmitting[27,31]. FMCW radars, by contrast, transmit a continuous, frequency-swept signal and measure beat frequency differences to determine range[32]. They can achieve very fine range resolution at short distances and essentially eliminate the close-in blind zone[27,28,32]. This makes FMCW radars well-suited for detecting nearby small obstacles that pulse radars might miss[33]. Modern solid-state marine radars often use FMCW waveforms[30,34] (sometimes called "broadband" radar in the marine market) to improve safety in tight navigation situations, since they can detect objects right up to the vessel's bow[33].

Marine radars most commonly operate in the X-band (~8–12 GHz) or S-band (~2–4 GHz) frequency ranges[29,35]. X-band radars, with their shorter ~3 cm wavelength, offer higher angular resolution and are more sensitive to small reflectors; indeed, the majority of marine navigational radars are X-band because the smaller antennas are practical for vessels and they provide better target resolution for small objects[36,37,38,39]. An X-band radar can detect smaller targets (and produce finer detail) thanks to the higher frequency and narrower beamwidth.[36] S-band radars, using a ~10 cm wavelength, have lower resolution but better propagation in rain, fog, and at long range.[36] They are often used on large ships or coastal surveillance systems for long-range coverage and for maintaining performance in heavy precipitation[36]. In practice, many ships carry both types: the X-band radar for high-resolution spotting

of small nearby targets, and the S-band radar for long-range situational awareness in all weather.[36] The complementary use of these bands ensures detection of small objects (with X-band) while still being able to operate in adverse weather that might attenuate X-band signals (using S-band)[36].

1.3.1.2 Detection challenges (sea clutter and target characteristics)

Detecting small targets on the sea surface is notoriously difficult due to environmental interference and target physics.[40,41,42] The ocean surface continuously generates radar reflections known as *sea clutter*, arising from waves and sea spray.[43,44] This clutter can often be of comparable or greater magnitude than the return from a small object, effectively burying the target in noise. Higher sea states (rougher seas with larger waves) dramatically increase clutter intensity and spatial variability, which in turn leads to a higher false alarm floor and can easily mask objects in wave troughs or among whitecaps. Small targets may also appear and disappear from radar as they move between wave peaks (line-of-sight intermittency) or as their orientation changes.[45,46] Moreover, many small maritime objects have inherently low RCS; for example, a swimmer's head barely above water or a piece of debris presents only a tiny reflective area to the radar.[47,48] The combination of low signal and high clutter means the signal-to-clutter ratio for such targets is extremely unfavorable[2].

Another consideration is the radar grazing angle. Maritime radars on vessels or low-flying aircraft view surface targets at low angles above the horizon. At low grazing angles, even moderate waves can significantly obstruct line-of-sight, and the reflective properties of the sea surface can cause multipath effects that weaken the direct return from a small target.[49,50,51,52,53] The sea surface itself, especially when wind-blown, produces random high-intensity spikes in the radar returns (from wave tops, splashes, etc.) which can either mimic a target or overwhelm it.[46, 40, 29, 3] These sporadic clutter spikes can lead to false detections and can even eclipse a real target's echo if they occur at the same range and angle. For instance, a radar operator might see momentary blips caused by breaking waves and have difficulty distinguishing them from a genuine small object bobbing in the water.

In summary, the maritime radar detection of small objects is limited by a mix of factors: low RCS of the targets, strong and dynamic sea clutter, shadowing by waves, and low viewing angles; all contributing to poor detectability and high false-alarm rates.

1.3.1.3 Countermeasures and processing techniques

Over decades, various radar signal processing techniques have been developed to improve small target detection in sea clutter.[54, 55, 56] A fundamental technique is the Constant False Alarm Rate (CFAR) detector, which adaptively adjusts the detection threshold based on the local clutter level[57]. In a CFAR scheme, the radar computes the mean or statistical distribution of the return power in a sliding range-angle window and sets a threshold such that the probability of false alarm is held constant (e.g., 10^{-6}). Any return that exceeds this local threshold is declared a target.[57] CFAR helps the radar dynamically adapt to varying sea states, for example, raising the threshold in heavy clutter and lowering it in calm conditions, rather than using a fixed threshold that would either miss targets or trigger too many false alarms. However, CFAR has limitations: if a small target is embedded in a strong clutter patch, the local threshold might be raised so high that the target is missed altogether (CFAR suffers in intense, non-homogeneous clutter environments).[58, 59] Conversely, in very light clutter, CFAR can become overly sensitive and still requires careful setting of false-alarm probability to avoid excessive false detections.[60,61,62]

Another important technique is Moving Target Indication (MTI) or Doppler filtering. [63,64] Many marine radars, especially modern solid-state units, can operate in pulse-Doppler mode or use coherent processing to measure the radial velocity of targets.[30, 65, 66] By exploiting the Doppler effect, the radar can distinguish moving targets from stationary or slow-moving clutter.[64, 67] Sea clutter from waves tends to have a characteristic velocity spectrum (related to wave motion), often concentrated around zero radial speed for simply drifting waves, with some spread due to wave surfaces moving toward or away from the radar.[68, 69, 70, 71] A swimmer or a small boat, even if slow, typically has a different radial

velocity signature than the random motion of waves.[72] Doppler processing can thus filter out returns that do not have a consistent velocity, effectively suppressing a large portion of sea clutter and highlighting real moving objects.[73, 74] In practice, applying MTI can greatly reduce false alarms from wave clutter and allows detection of small moving targets that would be lost in the clutter on a non-coherent radar.[75] The limitation here is that very slow or stationary objects (e.g., a nearly stationary person in water, or a piece of debris drifting at the same speed as the current) may also be filtered out as part of “clutter.”[76, 77] Additionally, the complex motions of sea clutter (especially in high sea states) can reduce the effectiveness of simple Doppler filters, since wave clutter is not truly stationary; hence, advanced algorithms like adaptive clutter maps or spectral subtraction are often needed.[73, 78, 79]

High-resolution radar techniques further aid small object detection.[80,81] Pulse compression (using chirped pulses or coded waveforms) can achieve fine range resolution while still using a powerful transmitted pulse, thereby improving the ability to separate a small target echo from adjacent clutter reflections.[82, 83] Narrow-beam antennas (possible at X-band and higher frequencies) improve angular resolution so that a small object occupies more distinct radar resolution cells, making it easier to detect.[84, 85]. Even polarization diversity has been explored; sea clutter and small targets can have different polarization scattering characteristics, and some studies leverage dual-polarized radar to discriminate small targets from clutter.[18, 86] These measures, however, cannot completely solve the problem; they only mitigate it to an extent.[56, 70]

Importantly, novel radar technologies are emerging to address the very small end of target detection.[87,88] For example, there are now high-frequency millimeter-wave marine radars (operating at K-band ~24 GHz or W-/E-band ~76 GHz) using FMCW waveforms that provide extremely high resolution at short ranges[89,32]. These systems can produce a detailed radar image of the environment around a vessel within a few hundred meters. They have demonstrated the capability to detect objects the size of a navigation buoy; and with tuning, even the size of a human or lifejacket in the water; by filtering out wave returns[90,91]. Some high-frequency radar can be configured to pick up a swimmer in the water

by adjusting its software to recognize smaller targets, although doing so requires careful control of interference from wave clutter[72,92]. The trade-off is that these high-frequency radars have limited range (hundreds of meters) and can be more affected by atmospheric attenuation (especially in rain or fog) compared to lower-frequency X- or S-band radars[93,94]. Thus, they complement rather than replace the traditional marine radar; a ship might use a 76 GHz radar for docking, maneuvering, and search-and-rescue operations in its immediate vicinity, while relying on X/S-band for long-range scanning.

In summary, marine radar provides the baseline for all-weather detection at sea, but it faces inherent difficulties in spotting small, low-RCS objects in clutter. Techniques like CFAR and Doppler processing are standard countermeasures that improve detection performance, yet small object detection remains challenging in high clutter or high sea state scenarios. These limitations motivate the exploration of augmenting radar with other sensing modalities, such as optical cameras, to help detect what radar might miss and vice versa.

1.3.2 Radar-Assisted Camera Labeling

On the visual side, advanced object detection algorithms have become standard tools for surveillance and autonomous systems.[95,96] In recent years, deep learning models, notably convolutional neural networks, have yielded state-of-the-art performance in detecting objects within camera imagery[97,98]. Two broad families of detectors exist: two-stage detectors like Faster R-CNN (which first propose candidate regions and then classify them) and one-stage detectors like YOLO (You Only Look Once) which directly regress bounding boxes and class probabilities in a single pass[5,99,100,101,102,103]. These methods have been successfully applied to detect ships, boats, and other marine objects in optical images[104,105]. For example, variants of Faster R-CNN and YOLO have been used in port surveillance and coastguard systems to identify vessels[106,107]. However, a critical challenge for deploying deep learning in maritime object detection is the *data requirement*[108,109]. These models typically need large,

diverse, annotated datasets to learn to detect objects reliably[110,111]. In the maritime domain, such datasets are scarce and costly to build[5,109,112]. Unlike common benchmarks (e.g., COCO or ImageNet) that have millions of everyday objects[111,113], maritime image datasets with small objects (e.g., a person overboard or a piece of debris) are few and far between[114,115]. Gathering thousands of images of, say, swimmers in open water, with ground truth annotations, is logistically difficult and potentially dangerous. Thus, data scarcity has hindered the full adoption of deep learning for small-object detection on the water; and, models trained on limited data may not generalize well to the wide range of ocean backgrounds and conditions.[109,114]

This is where radar-assisted camera labeling comes into play. The idea is to leverage the radar sensor as an adjunct to the camera not only for real-time detection, but also for generating training data. In a radar-assisted labeling approach, we use radar detections to propose candidate locations of objects in the visual data. Instead of a human annotator searching blindly through each image or video frame for a tiny object, the radar can direct the annotator's attention to specific regions. Practically, this could work as follows: whenever the radar detects a target (for instance, a radar return above a threshold at a certain range and bearing), that information is translated into the camera frame coordinates (given a calibrated setup). A *bounding box* or highlight is then automatically placed in the camera image at that location. The annotator simply checks if there is indeed an object visible in that region and adjusts the box if necessary. This turns the labeling task from a needle-in-haystack search to a targeted verification task. Over time, such a system could automatically accumulate a large number of labeled instances – especially of targets that are often missed in manual labeling, like low-contrast objects on a cluttered sea surface or objects present in poor lighting.

In essence, radar is used to **seed** the optical detector's training process with candidate examples. This concept is analogous to generating region proposals in computer vision, but here the proposals come from a different modality. Notably, some prior works have demonstrated the effectiveness of using one sensor's output to guide another; for example, in autonomous driving, LIDAR or radar have been used to generate

region proposals to assist camera-based detection algorithms[5]. Our context is similar: the radar acts as a proposal generator for the camera. The expected benefits include: (1) **Reduced human labeling effort** – since the system automatically flags likely object locations, human annotators spend less time per image, potentially an order-of-magnitude speedup in labeling rare events; (2) **Improved training data under tough conditions** – radar can detect objects even when they are invisible or hard to discern in the camera image (e.g., at night or in fog), so it can help collect positive examples that a purely manual or vision-based collection might miss. For instance, a person in dark clothing floating in twilight might be almost indiscernible in a video frame, but a radar may still get a return; with radar cueing, that frame can be labeled (perhaps noting it as a very low-visibility example) and included in training data, thereby improving the model's robustness to such conditions.

Using radar to assist vision-based detection also has potential for real-time fusion during operation. A fused **radar–camera detection system** could cross-verify detections: if the radar reports an object and the camera also sees something at that location, confidence in the detection should be higher. Conversely, if the camera detects an object but radar returns are absent, it might be a false optical detection (or something like a reflection on water surface), and vice versa for radar false alarms. Such cross-sensor verification can enhance reliability.

1.3.2.1 Challenges in radar-camera labeling and fusion:

Despite its promise, radar-assisted labeling and detection fusion comes with non-trivial challenges. A primary issue is ensuring the **alignment** of radar data with the camera image.[116,117] The radar provides detections in polar coordinates (range and bearing relative to the radar), whereas the camera image is a projective 2D plane.[6,118] To overlay radar detections onto the image, one needs an accurate extrinsic calibration (position and orientation of radar with respect to the camera) and synchronization in time[5,119,120]. Any calibration errors can lead to radar points projected slightly off from the true object location in the image.[116,119] For example, if the timing is off by even a fraction of a second, a

fast-moving object like a speedboat might appear at a different pixel location than where the radar indicated.[120,121] Furthermore, the radar and camera have very different *resolution* and coverage[122,123]. A radar might have an angular resolution of a few degrees and range resolution of several meters; a camera pixel might correspond to a few milliradians and a much smaller spatial patch at near distance[122,124]. This means a single radar detection might correspond to a relatively large region in the image or even multiple visually distinct objects (if, say, two objects are within one radar beam).[124] The radar also does not directly provide an object shape – it might just tell us "something is roughly here." [123,125] Thus, a radar-indicated ROI might not tightly fit the object in the image; some algorithmic steps are needed to refine the radar cue into a proper bounding box (possibly using the image gradients or an initial run of a vision detector on that region).

Additionally, **differences in modality** can cause mis-matches[126,127]. Radar detects based on reflectivity (metal objects or corner-like shapes reflect well; a person with a small cross-section or non-metallic debris might reflect poorly)[5,128,129]. A camera, on the other hand, detects objects based on contrast in color/texture against the background[130,131]. It's possible for radar to detect something that the camera effectively cannot see (e.g., a piece of wood barely protruding above water that is camouflaged to the eye but has an odd shape that radar picks up). In such cases, the radar-seeded "label" might appear to the human annotator as a box around nothing obvious; this could either lead to missed labels (if the human rejects it) or to the discovery of objects that would otherwise go unlabeled (if the human uses additional context or multiple frames to confirm the object). Managing these cases requires careful interface design for the annotator and possibly the integration of additional information (like infrared imagery or higher zoom) to confirm radar-only detections.

Another challenge is **false alarms and noise from radar**. Marine radars, as discussed, do produce occasional false plots due to sea spikes or interference. Feeding those directly into an annotation pipeline could introduce erroneous labels (ghost targets). It will be important to apply radar-based filtering (like requiring persistence over multiple scans or Doppler-consistency) to reduce false positives before using

radar outputs to label images. There is also the challenge of **multi-target association**; if multiple objects are present, associating each radar detection with the correct optical object (and maintaining that association over time) can be complex. Prior multi-sensor systems for maritime domain have used data association and tracking algorithms to fuse camera and radar detections. For instance, a system for an autonomous surface vessel might use radar to detect obstacles and a camera to classify them; ensuring the radar blip corresponds to the same physical object seen by the camera involves solving a data association problem, often via tracking filters or spatio-temporal gating.

In summary, radar-assisted camera labeling and detection is a promising approach to improve small object detection in maritime environments. The radar offers a means to greatly expand and enhance the training data available for deep learning models by providing automatic cues for object presence. It also offers a pathway to a **sensor-fusion detection system** where the strengths of radar (range, all-weather detection) and camera (detailed recognition) are combined. Successful implementation will require addressing alignment and calibration issues, handling the differing nature of radar vs. optical data, and filtering out spurious radar returns so they do not corrupt the training process. If these challenges are met, the outcome would be a more robust detection capability for small maritime objects and a semi-automated labeling tool that accelerates the development of future maritime computer vision systems.

References

1. **Guo, Z.-X., et al.** (2024). Small target detection in sea clutter using dominant clutter tree based on anomaly detection framework. *Signal Processing*, 219, 109399. <https://doi.org/10.1016/j.sigpro.2024.109399>
2. **Kunz, G. J., et al.** (2005). Detection of small targets in a marine environment using laser radar. In *Proc. SPIE 5885, Remote Sensing of the Coastal Oceanic Environment*, 58850W <https://doi.org/10.1117/12.614914>
3. **Sira, S. P., et al.** (2007). Adaptive waveform design for improved detection of low-RCS targets in heavy sea clutter. *IEEE Journal on Selected Topics in Signal Processing*, 1(1), 56–66. <https://doi.org/10.1109/JSTSP.2007.897048>
4. **Zhang, X.-J., et al.** (2024). Long-time adaptive coherent detection of small targets in sea clutter by fast inversion algorithm of block tridiagonal speckle covariance matrices. *Signal Processing*, 217, 109326. <https://doi.org/10.1016/j.sigpro.2023.109326>
5. **Zhao, C., et al.** (2024). Deep Learning-Based Object Detection in Maritime Unmanned Aerial Vehicle Imagery: Review and Experimental Comparisons. *Engineering Applications of Artificial Intelligence*, 128, 107513. <https://doi.org/10.1016/j.engappai.2023.107513>
6. **Shi, K., et al.** (2024). *Radar and Camera Fusion for Object Detection and Tracking: A Comprehensive Survey*. arXiv:2410.19872 <https://doi.org/10.48550/arXiv.2410.19872>
7. **Jin, B., et al.** (2017). A fast scan-to-scan integration algorithm for small target in sea clutter. *Electronics Letters*, 53(15), 1070–1071. <https://doi.org/10.1049/el.2017.1312>
8. **Becktor, J. B., et al.** (2022). Re-annotation of training samples for robust maritime object detection. *Machine Learning with Applications*, 10, 100411. <https://doi.org/10.1016/j.mlwa.2022.100411>
9. **Becktor, J. B., et al.** (2022). Bolstering maritime object detection with synthetic data. In *Proc. 14th IFAC Conference on Control Applications in Marine Systems, Robotics and Vehicles, IFAC-PapersOnLine*, 55(31), 64–69. <https://doi.org/10.1016/j.ifacol.2022.10.410>
10. **Zust, L., & Kristan, M.** (2022). Learning Maritime Obstacle Detection from Weak Annotations by Scaffolding. In *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 1888–1897. <https://doi.org/10.1109/wacv51458.2022.00195>
11. **Goodwin, M., et al.** (2022). Unlocking the potential of deep learning for marine ecology: overview, applications, and outlook. *ICES Journal of Marine Science*, 79(2), 319–336. <https://doi.org/10.1093/icesjms/fsab255>
12. **Tran, M., et al.** (2024). SafeSea: Synthetic data generation for adverse & low probability maritime conditions. In *Proc. IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW 2024)* (pp. 821–829). <https://doi.org/10.1109/WACVW60836.2024.00094>
13. **Hashmani, M. A., & Umair, M.** (2022). A novel visual-range sea image dataset for sea horizon line detection in changing maritime scenes. *Journal of Marine Science and Engineering*, 10(2), 193. <https://doi.org/10.3390/jmse10020193>
14. **Raza, M., et al.** (2022). SimuShips – A high resolution simulation dataset for ship detection with precise annotations. In *Proc. OCEANS 2022, Hampton Roads* (pp. 1–5). <https://doi.org/10.1109/OCEANS47191.2022.9977182>
15. **Ochal, M., et al.** (2020). A comparison of few-shot learning methods for underwater optical and sonar image classification. In *Proc. IEEE Global OCEANS 2020: Singapore – U.S. Gulf Coast* (pp. 1–10). <https://doi.org/10.1109/IEEECONF38699.2020.9389475>
16. **Ratner, A., et al.** (2017). Snorkel: Rapid training data creation with weak supervision. *Proceedings of the VLDB Endowment*, 11(3), 269–282. <https://doi.org/10.14778/3157794.3157797>
17. **Feng, J., et al.** (2024). Adaptive adversarial self-training for semi-supervised object detection in complex maritime scenes. *Mathematics*, 12(15), 2348. <https://doi.org/10.3390/math12152348>

18. **Fan, Y., et al.** (2024). Weak Target Detection Based on Full-Polarization Scattering Features under Sea Clutter Background. *Remote Sensing*, 16(16), 2987.
<https://doi.org/10.3390/rs16162987>
19. **Bounaceur, H., et al.** (2022). Analysis of Small Sea-Surface Targets Detection Performance According to Airborne Radar Parameters in Abnormal Weather Environments. *Sensors*, 22(9), 3263. <https://doi.org/10.3390/s22093263>
20. **Ousbome, Jeffrey J., et al.** "A periscope detection radar." *Johns Hopkins APL Technical Digest* 18, no. 1 (1997): 125-133.
<https://secwww.jhuapl.edu/techdigest/Content/techdigest/pdf/V18-N01/18-01-Ousbourn.pdf>
21. **Sheeny, M.** (2020). All-Weather Object Recognition Using Radar and Infrared Sensing. *arXiv:2010.16285* <https://doi.org/10.48550/arXiv.2010.16285>
22. **Narula, L., et al.** (2022). All-Weather, Sub-50-cm, Radar-Inertial Positioning. *Field Robotics*, 2, 525–556. <https://doi.org/10.55417/fr.2022019>
23. **Reina, G., et al.** (2011). Radar-Based Perception for Autonomous Outdoor Vehicles. *Journal of Field Robotics*, 28(6), 894–913. <https://doi.org/10.1002/rob.20393>
24. **Ryde, J., & Hillier, N.** (2009). Performance of Laser and Radar Ranging Devices in Adverse Environmental Conditions. *Journal of Field Robotics*, 26(9), 712–727.
<https://doi.org/10.1002/rob.20310>
25. **Nabati, R., & Qi, H.** (2019). RRPN: Radar Region Proposal Network for Object Detection in Autonomous Vehicles. In *Proceedings of the IEEE International Conference on Image Processing (ICIP)* (pp. 3093–3097). <https://doi.org/10.1109/ICIP.2019.8803392>
26. **Dong, X., et al.** (2021). Radar Camera Fusion via Representation Learning in Autonomous Driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2021)* (pp. 1672–1681).
<https://doi.org/10.1109/CVPRW53098.2021.00316>
27. **Rudys, S., et al.** (2022). Investigation of UAV Detection by Different Solid-State Marine Radars. *Electronics*, 11(16), 2502. <https://doi.org/10.3390/electronics11162502>
28. **Zhao, H., et al.** (2021). High-Frequency Radar Cross Section of Ocean Surface for an FMICW Source. *Journal of Marine Science and Engineering*, 9(4), 427.
<https://doi.org/10.3390/jmse9040427>
29. **Williams, P. D. L.** (1993). Marine Radar: A Selective Review of R&D and a Way Ahead. *The Journal of Navigation*, 46(3), 442–447. <https://doi.org/10.1017/S0373463300011899>
30. **Galati, G., et al.** (2016). Compatibility problems related with pulse-compression, solid-state marine radars. *IET Radar, Sonar & Navigation*, 10(4), 791–797.
<https://doi.org/10.1049/iet-rsn.2015.0400>
31. **Salazar Aquino, C. M., et al.** (2021). Progressive Pulse Compression: A Novel Technique for Blind Range Recovery for Solid-State Radars. *Journal of Atmospheric and Oceanic Technology*, 38(9), 1599–1611. <https://doi.org/10.1175/JTECH-D-20-0164.1>
32. **Pirkani, A., et al.** (2024). 77 GHz FMCW Imaging Radar for Low Observable and Small Marine Target Detection in Dynamic Sea Conditions Based on Combined MIMO and DBS. *IEEE Transactions on Radar Systems*. <https://doi.org/10.1109/TRS.2024.3400694>
33. **Bogner, M., et al.** (2024). Utilizing 1D FMCW Radar Data for Distance Estimation to Port Infrastructure. In *Proc. 27th International Conference on Information Fusion (FUSION 2024)*.
<https://doi.org/10.23919/FUSION59988.2024.10706500>
34. **Rudys, S., et al.** (2021). Functionality Investigation of the UAV Arranged FMCW Solid-State Marine Radar. *Journal of Marine Science and Engineering*, 9(8), 887.
<https://doi.org/10.3390/jmse9080887>
35. **Nieto-Borge, J. C., et al.** (2008). Signal-to-noise ratio analysis to estimate ocean wave heights from X-band marine radar image time series. *IET Radar, Sonar & Navigation*, 2(1), 35–41.
<https://doi.org/10.1049/iet-rsn:20070027>

36. Rutkowski, G. (2020). Practical aspects of using radar in maritime navigation with analysis of its potential errors and limitations. *Scientific Journal of Gdynia Maritime University*, (113), 17–28. <https://doi.org/10.26408/113.02>
37. Horstmann, J., et al. (2015). Wind, wave, and current retrieval utilizing X-band marine radars. In Y. Liu, H. Kerkerling, & R. H. Weisberg (Eds.), *Coastal Ocean Observing Systems* (pp. 281–304). Elsevier. <https://doi.org/10.1016/B978-0-12-802022-7.00016-X>
38. Vivone, G., et al. (2015). Extended target tracking using joint probabilistic data association filter on X-band radar data. In Proc. 2015 *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Milan, Italy, 26–31 July 2015. IEEE. <https://doi.org/10.1109/IGARSS.2015.7326971>
39. Arturi, D., et al. (2016). X band radar target tracking in marine environment: A comparison of different algorithms in a real scenario. In Proc. 2016 *10th European Conference on Antennas and Propagation (EuCAP)*, Davos, Switzerland, 10–15 April 2016. IEEE. <https://doi.org/10.1109/EuCAP.2016.7481294>
40. Xie, J., & Xu, X. (2022). Phase-feature-based detection of small targets in sea clutter. *IEEE Geoscience and Remote Sensing Letters*, 19, Article 3507405. <https://doi.org/10.1109/LGRS.2021.3093620>
41. Chen, Y., et al. (2023). Small Target Detection on Sea Surface based on Feature-Weighted Support Vector Machine. In Proc. 2023 *8th International Conference on Signal and Image Processing (ICSIP)*, Wuxi, China, 08–10 July 2023. IEEE. <https://doi.org/10.1109/ICSIP57908.2023.10271047>
42. Xu, S., et al. (2020). Status and prospects of feature-based detection methods for floating targets on the sea surface. *Journal of Radars*, 9(4), 684–714. <https://doi.org/10.12000/JR20084>
43. Shao, Z., et al. (2020). Extraction of a target in sea clutter via signal decomposition. *Science China Information Sciences*, 63, 129301. <https://doi.org/10.1007/s11432-018-9859-4>
44. Das, N., & Hossain, M. S. (2023). Investigation of the impact of sea conditions on the sea surface reflectivity in maritime radar sea clutter modeling. In Proc. 3rd *International Conference on Electrical, Computer and Communication Engineering (ECCE 2023)* (pp. 1–4). IEEE. <https://doi.org/10.1109/ECCE57851.2023.10101598>
45. Dong, Y., et al. (2024). A Dynamic False Alarm Rate Control Method for Small Target Detection in Non-Stationary Sea Clutter. *Journal of Marine Science and Engineering*, 12(10), 1770. <https://doi.org/10.3390/jmse12101770>
46. Shi, S.-N., et al. (2022). Small Target Detection Based on Noncoherent Radial Velocity Spectrum of High-Resolution Sea Clutter. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15, 8719–8733. <https://doi.org/10.1109/JSTARS.2022.3213420>
47. Tian, C., et al. (2016). Small target detection for solid-state marine radar. In Proc. 2016 *CIE International Conference on Radar (RADAR)*, Guangzhou, China, 10–13 October 2016. IEEE. <https://doi.org/10.1109/RADAR.2016.8059302>
48. Frazer, G. J., & Williams, C. G. (2020). HF RCS of Small Boat Displacement in the Ocean. In Proc. 2020 *IEEE Radar Conference (RadarConf20)*, Florence, Italy, 21–25 September 2020. IEEE. <https://doi.org/10.1109/RadarConf2043947.2020.9266341>
49. Tao, M., et al. (2024). Low-angle tracking of shipborne radar. In Proc. 2024 *IEEE 6th International Conference on Power, Intelligent Computing and Systems (ICPICS)*, Shenyang, China, 26–28 July 2024. IEEE. <https://doi.org/10.1109/ICPICS62053.2024.10796622>
50. Xia, Y., et al. (2015). Target Detection in Low Grazing Angle with Adaptive OFDM Radar. *International Journal of Antennas and Propagation*, 2015, Article ID 147368, 12 pages. <https://doi.org/10.1155/2015/147368>
51. Daniel, L., et al. (2015). Target visibility estimation in a buoy mounted maritime forward scatter radar. In Proc. 2015 *Radar Conference (RadarCon)*, Vol. 2015, pp. 755–760. IEEE. <https://doi.org/10.1109/RADAR.2015.7131097>

52. **Gregers-Hansen, V., & Mital, R.** (2009). An empirical sea clutter model for low grazing angles. In *Proc. 2009 IEEE Radar Conference*, Pasadena, CA, USA, 04–08 May 2009. IEEE. <https://doi.org/10.1109/RADAR.2009.4977006>
53. **Trizna, D. B.** (1997). A model for Brewster angle damping and multipath effects on the microwave radar sea echo at low grazing angles. *IEEE Transactions on Geoscience and Remote Sensing*, 35(5), 1232–1243. <https://doi.org/10.1109/36.628790>
54. **Singh, J. P.** (2005). Evolution of the radar target tracking algorithms: A move towards knowledge based multi-sensor adaptive processing. In *Proc. 1st IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing*, Puerto Vallarta, Mexico, 13–15 December 2005. IEEE. <https://doi.org/10.1109/CAMAP.2005.1574178>
55. **Davis, M. E.** (2015). A history of battlefield surveillance radar. In *Proc. 2015 IEEE Radar Conference (RadarCon)*, Arlington, VA, USA, 10–15 May 2015. IEEE. <https://doi.org/10.1109/RADAR.2015.7131204>
56. **Navya, R., & Devaraju, R.** (2021). A systematic survey on radar target detection techniques in sea clutter background. In *Smart Computing Techniques and Applications* (Smart Innovation, Systems and Technologies, Vol. 224, pp. 327–336). Springer. https://doi.org/10.1007/978-981-16-1502-3_33
57. **Finn, H. M., & Johnson, R. S.** (1968). Adaptive detection mode with threshold control as a function of spatially sampled clutter-level estimates. *RCA Rev.*, 29(3), 414–465.
58. **Tabet, L.** (2024). Analyses of the trimmed-mean switching CFAR detector in non-homogenous Gaussian clutter. *Studies in Engineering and Exact Sciences*, 5(3), e12900. <https://doi.org/10.54021/seesv5n3-111>
59. **Colgin, R. C.** (1998). Description and analysis of a Bayesian CFAR radar signal processor in a nonhomogeneous clutter background. In *Proc. 1998 IEEE Radar Conference (RadarCon '98): Challenges in Radar Systems and Solutions* (pp. 3–8). IEEE. <https://doi.org/10.1109/NRC.1998.677968>
60. **Shin, J. H., & Choi, Y.** (2020). Robust control for the detection threshold of CFAR process in cluttered environments. *Sensors*, 20(14), 3904. <https://doi.org/10.3390/s20143904>
61. **Rohling, H.** (1983). Radar CFAR thresholding in clutter and multiple target situations. *IEEE Transactions on Aerospace and Electronic Systems*, 19(4), 608–621. <https://doi.org/10.1109/TAES.1983.309350>
62. **Baadeche, M., & Soltani, F.** (2013). Performance comparison of some CFAR detectors in homogenous and non-homogenous clutter. In *Proc. 2013 IEEE International Conference on Signal and Image Processing Applications* (pp. 101–105). IEEE. <https://doi.org/10.1109/ICSIPA.2013.6707985>
63. **Xu, J., et al.** (2011). Radon-Fourier Transform for Radar Target Detection, I: Generalized Doppler Filter Bank. *IEEE Transactions on Aerospace and Electronic Systems*, 47(2), 1186–1202. <https://doi.org/10.1109/TAES.2011.5751251>
64. **Barlow, E. J.** (1949). Doppler Radar. *Proceedings of the IRE*, 37(4), 340–355. <https://doi.org/10.1109/JRPROC.1949.231638>
65. **Abdelbagi, H., et al.** (2015). FPGA-based coherent Doppler processor for marine radar applications. In *Proc. 2015 National Aerospace and Electronics Conference (NAECON)* (pp. 58–69). IEEE. <https://doi.org/10.1109/NAECON.2015.7443040>
66. **Lyzenga, D. R.** (2017). Polar Fourier Transform Processing of Marine Radar Signals. *Journal of Atmospheric and Oceanic Technology*, 34(2), 347–354. <https://doi.org/10.1175/JTECH-D-16-0158.1>
67. **Chen, V. C.** (2002). Radar detection of multiple moving targets in clutter using time-frequency Radon transform. In *Signal and Data Processing of Small Targets 2002* (Vol. 4728, pp. 48–59). SPIE. <https://doi.org/10.1117/12.478534>
68. **Lv, M., & Zhou, C.** (2019). Study on sea clutter suppression methods based on a realistic radar dataset. *Remote Sensing*, 11(23), 2721. <https://doi.org/10.3390/rs11232721>

69. **Bocquet, S.** (2022). Analysis and simulation of low grazing angle X-Band coherent radar sea clutter using memoryless nonlinear transformations. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1–13. <https://doi.org/10.1109/TGRS.2022.3153632>
70. **Watts, S., & Rosenberg, L.** (2022). Challenges in radar sea clutter modelling. *IET Radar, Sonar & Navigation*, 16(9), 1403–1414. <https://doi.org/10.1049/rsn2.12272>
71. **Watts, S., et al.** (2016). Doppler spectra of medium grazing angle sea clutter; part 2: model assessment and simulation. *IET Radar, Sonar & Navigation*, 10(1), 32–42. <https://doi.org/10.1049/iet-rsn.2015.0149>
72. **Park, J., et al.** (2016). Micro-Doppler based classification of human aquatic activities via transfer learning of convolutional neural networks. *Sensors*, 16(12), 1990, 2016. <https://doi.org/10.3390/s16121990>
73. **Brennan, L. E., & Reed, L. S.** (1973). Theory of adaptive radar. *IEEE Transactions on Aerospace and Electronic Systems*, AES-9(2), 237–252. <https://doi.org/10.1109/TAES.1973.309792>
74. **Ward, K. D., et al.** (1990). Maritime surveillance radar. Part 1: Radar scattering from the ocean surface. *IEE Proceedings F: Radar and Signal Processing*, 137(2), 51. <https://doi.org/10.1049/ip-f-2.1990.0009>
75. **Zhang, W., et al.** (2017). Optimization of Adaptive MTI Filter. *International Journal of Communications, Network and System Sciences*, 10(8), 206–217. <https://doi.org/10.4236/ijcns.2017.108B022>
76. **Kirubarajan, T., & Bar-Shalom, Y.** (2000). Tracking evasive move-stop-move targets with a MTI radar using a VS-IMM estimator. In O. E. Drummond (Ed.), *Signal and Data Processing of Small Targets 2000* (Vol. 4048, pp. 236–246). SPIE. <https://doi.org/10.1117/12.391980>
77. **Bickel, D. L., & Doerry, A. W.** (2019). On minimum detectable velocity. In K. I. Ranney & A. Doerry (Eds.), *Radar Sensor Technology XXIII* (p. 23). SPIE. <https://doi.org/10.1117/12.2518121>
78. **Verrazzani, L., et al.** (2000). Performance analysis of two adaptive radar detectors against non-Gaussian real sea clutter data. *IEEE Transactions on Aerospace and Electronic Systems*, 36(4), 1429–1439. <https://doi.org/10.1109/7.892695>
79. **Xu, S.-W., et al.** (2016). Combined adaptive normalized matched filter detection of moving target in sea clutter. *Circuits, Systems, and Signal Processing*, 36(6), 2360–2383. <https://doi.org/10.1007/s00034-016-0413-5>
80. **Li, J., et al.** (2021). Towards small target recognition with photonics-based high resolution radar range profiles. *Optics Express*, 29(20), 31574. <https://doi.org/10.1364/OE.439610>
81. **Hwang, W., et al.** (2023). Clutter cancellation methods for small target detection using high-resolution W-band radar. *Sensors*, 23(17), 7557. <https://doi.org/10.3390/s23177557>
82. **Suryanarayana, E. V., et al.** (2019). Pulse compression waveforms: Applications to radar systems. In *Advances in Decision Sciences, Image Processing, Security and Computer Vision* (pp. 61–68). Springer International Publishing. https://doi.org/10.1007/978-3-030-24318-0_8
83. **Levanon, N., & Cohen, I.** (2024). Waveforms search for noncoherent pulse compression. *IEEE Aerospace and Electronic Systems Magazine*, 39(4), 32–40. <https://doi.org/10.1109/MAES.2024.3365662>
84. **Long, T., et al.** (2019). Advanced technology of high-resolution radar: Target detection, tracking, imaging, and recognition. *Science China Information Sciences*, 62(4). <https://doi.org/10.1007/s11432-018-9811-0>
85. **Anim, K., et al.** (2021). Compact slotted waveguide antenna array using staircase model of tapered dielectric-inset guide for shipboard marine radar. *Sensors*, 21(14), 4745. <https://doi.org/10.3390/s21144745>
86. **Kim, J. E., et al.** (2014). Target detection using radar in heavy sea clutter by polarimetric analysis and neural network. In *Advances in Neural Networks – ISNN 2014* (pp. 479–488). Springer International Publishing. https://doi.org/10.1007/978-3-319-12436-0_53

87. **Chu, T., et al.** (2025). Intelligent Detection of Low–Slow–Small Targets Based on Passive Radar. *Remote Sensing*, 17(6), 961. <https://doi.org/10.3390/rs17060961>
88. **Cai, L., et al.** (2023). A Software-Defined Radar for Low-Altitude Slow-Moving Small Targets Detection Using Transmit Beam Control. *Remote Sensing*, 15(13), 3371. <https://doi.org/10.3390/rs15133371>
89. **Gennarelli, G., et al.** (2022). 24 GHz FMCW MIMO Radar for Marine Target Localization: A Feasibility Study. *IEEE Access*, 10, 68240–68256. <https://doi.org/10.1109/ACCESS.2022.3186052>
90. **Pirkani, A., et al.** (2024). 77 GHz FMCW Imaging Radar for Low Observable and Small Marine Target Detection in Dynamic Sea Conditions Based on Combined MIMO and DBS. *IEEE Transactions on Radar Systems*, 2, 517–534. <https://doi.org/10.1109/TRS.2024.3400694>
91. **Li, Y., et al.** (2019). SVM-Based Sea-Surface Small Target Detection: A False-Alarm-Rate-Controllable Approach. *IEEE Geoscience and Remote Sensing Letters*, 16(8), 1225–1229. <https://doi.org/10.1109/LGRS.2019.2894385>
92. **Zhou, Y., et al.** (2024). Radar-Based Swimming Activity Recognition with Temporal Dynamic Convolution and Spectral Data Augmentation. *IEEE Sensors Journal*, 1–1. <https://doi.org/10.1109/JSEN.2024.3459085>
93. **Bertoldo, S., et al.** (2018). On the Use of a 77 GHz Automotive Radar as a Microwave Rain Gauge. *Engineering, Technology & Applied Science Research*, 8(1), 2356–2360. <https://doi.org/10.48084/etasr.1755>
94. **Myagkov, A., et al.** (2020). Evaluation of the Reflectivity Calibration of W-Band Radars Based on Observations in Rain. *Atmospheric Measurement Techniques*, 13(11), 5799–5825. <https://doi.org/10.5194/amt-13-5799-2020>
95. **El-Alami, A., et al.** (2024). A Review of Object Detection Approaches for Traffic Surveillance Systems. *International Journal of Electrical and Computer Engineering (IJECE)*, 14(5), 5221–5233. <https://doi.org/10.11591/ijece.v14i5.pp5221-5233>
96. **Garg, H., et al.** (2023). A Review of Object Detection Algorithms for Autonomous Vehicles: Trends and Developments. In *Proceedings of the 2023 5th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*, 1173–1181. <https://doi.org/10.1109/ICAC3N60023.2023.10541773>
97. **Zhao, X., et al.** (2024). A review of convolutional neural networks in computer vision. *Artificial Intelligence Review*, 57(4). <https://doi.org/10.1007/s10462-024-10721-6>
98. **Noor, M. H. M., & Ige, A. O.** (2025). A Survey on State-of-the-art Deep Learning Applications and Challenges. arXiv:2403.17561. <https://doi.org/10.48550/arXiv.2403.17561>
99. **Ren, S., et al.** (2017). Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(6), 1137–1149. <https://doi.org/10.1109/TPAMI.2016.2577031>
100. **Girshick, R.** (2015). Fast R-CNN. In *2015 IEEE International Conference on Computer Vision (ICCV)*. <https://doi.org/10.1109/ICCV.2015.169>
101. **Redmon, J., et al.** (2016). You Only Look Once: Unified, Real-Time Object Detection. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 779–788. <https://doi.org/10.1109/CVPR.2016.91>
102. **Liu, W., et al.** (2016). SSD: Single Shot MultiBox Detector. In *Computer Vision – ECCV 2016*, 21–37. https://doi.org/10.1007/978-3-319-46448-0_2
103. **Lin, T.-Y., et al.** (2017). Focal Loss for Dense Object Detection. In *2017 IEEE International Conference on Computer Vision (ICCV)*. <https://doi.org/10.1109/ICCV.2017.324>
104. **Zhai, M., et al.** (2019). Ship Detection Based on Faster R-CNN Network in Optical Remote Sensing Images. In *Proceedings of the 2019 Chinese Intelligent Automation Conference*, 22–31. https://doi.org/10.1007/978-981-32-9050-1_3

105. **Fang, Z., et al.** (2024). YOLO-RSA: A Multiscale Ship Detection Algorithm Based on Optical Remote Sensing Image. *Journal of Marine Science and Engineering*, 12(4), 603. <https://doi.org/10.3390/jmse12040603>
106. **Qi, L., et al.** (2019). Ship Target Detection Algorithm Based on Improved Faster R-CNN. *Electronics*, 8(9), 959. <https://doi.org/10.3390/electronics8090959>
107. **Folarin, A., et al.** (2024). Real Time Vessel Detection Model Using Deep Learning Algorithms for Controlling a Barrier System. *Journal of Marine Science and Engineering*, 12(8), 1363. <https://doi.org/10.3390/jmse12081363>
108. **Zhang, R., et al.** (2021). Survey on Deep Learning-Based Marine Object Detection. *Journal of Advanced Transportation*, 2021, 1–18. <https://doi.org/10.1155/2021/5808206>
109. **Moosbauer, S., et al.** (2019). A Benchmark for Deep Learning Based Object Detection in Maritime Environments. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 916–925. <https://doi.org/10.1109/CVPRW.2019.00121>
110. **Miri Rekavandi, A., et al.** (2025). A Guide to Image- and Video-Based Small Object Detection Using Deep Learning: Case Study of Maritime Surveillance. *IEEE Transactions on Intelligent Transportation Systems*, 26(3), 2851–2879. <https://doi.org/10.1109/TITS.2025.3530678>
111. **Lin, T.-Y., et al.** (2014). Microsoft COCO: Common Objects in Context. In *Computer Vision – ECCV 2014*, 740–755. https://doi.org/10.1007/978-3-319-10602-1_48
112. **Ahlers, D., et al.** (2023). Shore Livecams: A Maritime Dataset for Deep Learning based Object Detection. In *2023 17th International Conference on Signal-Image Technology & Internet-Based Systems (SITIS)*, 138–144. <https://doi.org/10.1109/SITIS61268.2023.00029>
113. **Russakovsky, O., et al.** (2015). ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision*, 115(3), 211–252. <https://doi.org/10.1007/s11263-015-0816-y>
114. **Miri Rekavandi, A., et al.** (2025). A Guide to Image- and Video-Based Small Object Detection Using Deep Learning: Case Study of Maritime Surveillance. *IEEE Transactions on Intelligent Transportation Systems*, 26(3), 2851–2879. <https://doi.org/10.1109/TITS.2025.3530678>
115. **Choi, J., et al.** (2024). PoLaRIS Dataset: A Maritime Object Detection and Tracking Dataset in Pohang Canal. arXiv:2412.06192. <https://doi.org/10.48550/arXiv.2412.06192>
116. **Liu, X., et al.** (2025). Targetless Radar–Camera Extrinsic Parameter Calibration Using Track-to-Track Association. *Sensors*, 25(3), 949. <https://doi.org/10.3390/s25030949>
117. **Cheng, L., & Cao, S.** (2023). Online Targetless Radar-Camera Extrinsic Calibration Based on the Common Features of Radar and Camera. In *NAECON 2023 – IEEE National Aerospace and Electronics Conference*, 294–299. <https://doi.org/10.1109/NAECON58068.2023.10366051>
118. **Lin, Z., et al.** (2024). RCBEVDet: Radar-Camera Fusion in Bird’s Eye View for 3D Object Detection. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 14928–14937. <https://doi.org/10.1109/CVPR52733.2024.01414>
119. **Domhof, J., et al.** (2019). An Extrinsic Calibration Tool for Radar, Camera and Lidar. In *2019 International Conference on Robotics and Automation (ICRA)*. <https://doi.org/10.1109/ICRA.2019.8794186>
120. **Jellum, E. R., et al.** (2022). The Syncline Model – Analyzing the Impact of Time Synchronization in Sensor Fusion. In *2022 IEEE Conference on Control Technology and Applications (CCTA)*, 1446–1453. <https://doi.org/10.1109/CCTA49430.2022.9966179>
121. **Lee, S., et al.** (2024). Impact Analysis of Time Synchronization Error in Airborne Target Tracking Using a Heterogeneous Sensor Network. *Drones*, 8(5), 167. <https://doi.org/10.3390/drones8050167>
122. **Han, Z., et al.** (2024). 4D Millimeter-Wave Radar in Autonomous Driving: A Survey. arXiv:2306.04242. <https://doi.org/10.48550/arXiv.2306.04242>

123. **Ogunrinde, I., & Bernadin, S.** (2023). Deep Camera–Radar Fusion with an Attention Framework for Autonomous Vehicle Vision in Foggy Weather Conditions. *Sensors*, 23(14), 6255. <https://doi.org/10.3390/s23146255>
124. **Sun, R., et al.** (2023). A Millimeter-Wave Automotive Radar with High Angular Resolution for Identification of Closely Spaced On-Road Obstacles. *Scientific Reports*, 13(1). <https://doi.org/10.1038/s41598-023-30406-4>
125. **Sun, H., et al.** (2025). CaRaFFusion: Improving 2D Semantic Segmentation With Camera-Radar Point Cloud Fusion and Zero-Shot Image Inpainting. *IEEE Robotics and Automation Letters*, 1–8. <https://doi.org/10.1109/LRA.2025.3572418>
126. **Lin, Z., et al.** (2024). RCBEVDet++: Toward High-accuracy Radar-Camera Fusion 3D Perception Network. arXiv:2409.04979. <https://doi.org/10.48550/arXiv.2409.04979>
127. **Liu, X., et al.** (2024). Camera–Radar Fusion with Modality Interaction and Radar Gaussian Expansion for 3D Object Detection. *Cyborg and Bionic Systems*, 5. <https://doi.org/10.34133/cbsystems.0079>
128. **Yamada, N., et al.** (2005). Radar Cross Section for Pedestrian in 76 GHz Band. In *2005 European Microwave Conference*, 4 pp. – 1018. <https://doi.org/10.1109/EUMC.2005.1610101>
129. **Buchberger, C., et al.** (2019). Dielectric Corner Reflectors for mmWave Applications. *Advances in Radio Science*, 17, 197–203. <https://doi.org/10.5194/ars-17-197-2019>
130. **Zhang, L., et al.** (2016). Unified Saliency Detection Model Using Color and Texture Features. *PLOS ONE*, 11(2), e0149328. <https://doi.org/10.1371/journal.pone.0149328>
131. **Crandall, D., & Luo, J.** (2004). Robust Color Object Detection Using Spatial-Color Joint Probability Functions. In *Proceedings of the 2004 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2004)*, 379–385. <https://doi.org/10.1109/CVPR.2004.1315057>